lib -> core development files. lib is starting / core development folder

test -> unit/ widget _test.
assets -> Resources / Images / fonts / files / videos / Static Resources

platform specific folder
① android | windows
② ios -> | macos
③ web | linux

configuration files -> dependencies / assets declaration
↳ pubspec.yaml -> project Name, config, environment, dependencies, assets declaration

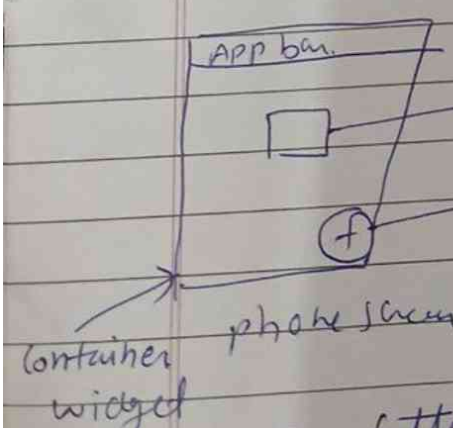
locked ← pubspec.lock
version of dependencies
analysis_option.yaml

id generated files

flutter commands
① flutter doctor
② flutter create Applicationname
③ cd Applicationname ← Android studio / VSCode
my-first-Project

flutter run -d chrom

widgets -> it is fundamental concept. it is basic building blocks



Text widget
Image widget
Button widget
] display

↳ A container widget is like box that can hold other widget

In flutter everything is widget, our entire app is just widgets inside widgets inside more widgets

Types

- State less widgets → no change static
- State ful widgets → can change

A stateless widget display information that doesnot change
↳ A text always show same message
↳ An image that never changes

] logo | text value by code

A statefull widget can change their appearance based on user interaction

traffic light → Red, green, blue } not static

The root of everything is <u>material App</u>

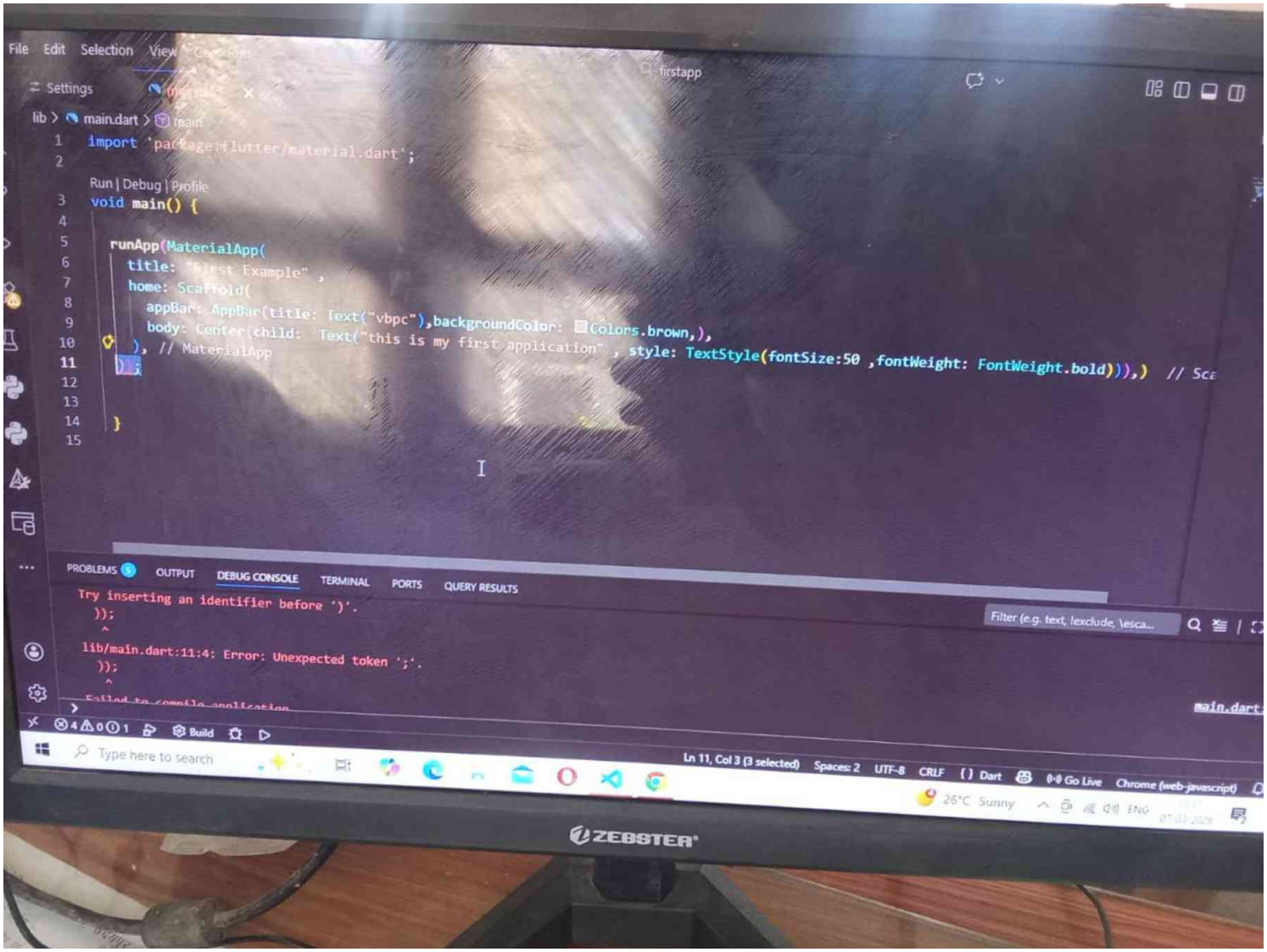
material App → set up meterial Design is modern Android look and feel that manage nevigation and provide app-wide setting — UI → Theame (color), button look

<u>Widget tree</u> → widgets are organized in something called widget tree.

materialApp (root
↳ home : MyHomePage
↳ Scaffold
→ AppBar
→ Body
↳ other widgets

```
void main ()
{
function ——— runApp (MaterialApp (
                                  (
widget ——— title: "Hello world",
           home: Scaffold (
                 body: Text ('Hello world')),
                                  ));
}
```

import 'package:flutter/material.dart';


File Edit Selection View
Settings
firstapp
lib > main.dart > main
import 'package:flutter/material.dart';
Run | Debug | Profile
void main() {
runApp(MaterialApp(
title: "First Example" ,
home: Scaffold(
appBar: AppBar(title: Text("vbpc"),backgroundColor: Colors.brown,),
body: Center(child: Text("this is my first application" , style: TextStyle(fontSize:50 ,fontWeight: FontWeight.bold))),)
), // MaterialApp
}
PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
Try inserting an identifier before ')'.
));
lib/main.dart:11:4: Error: Unexpected token ';'.
));
Build
Type here to search
Ln 11, Col 3 (3 selected) Spaces: 2 UTF-8 CRLF {} Dart Go Live Chrome (web-javascript)
26°C Sunny
ENG
ZEBSTER

Swym | Jult

Header

name

Section | article

article

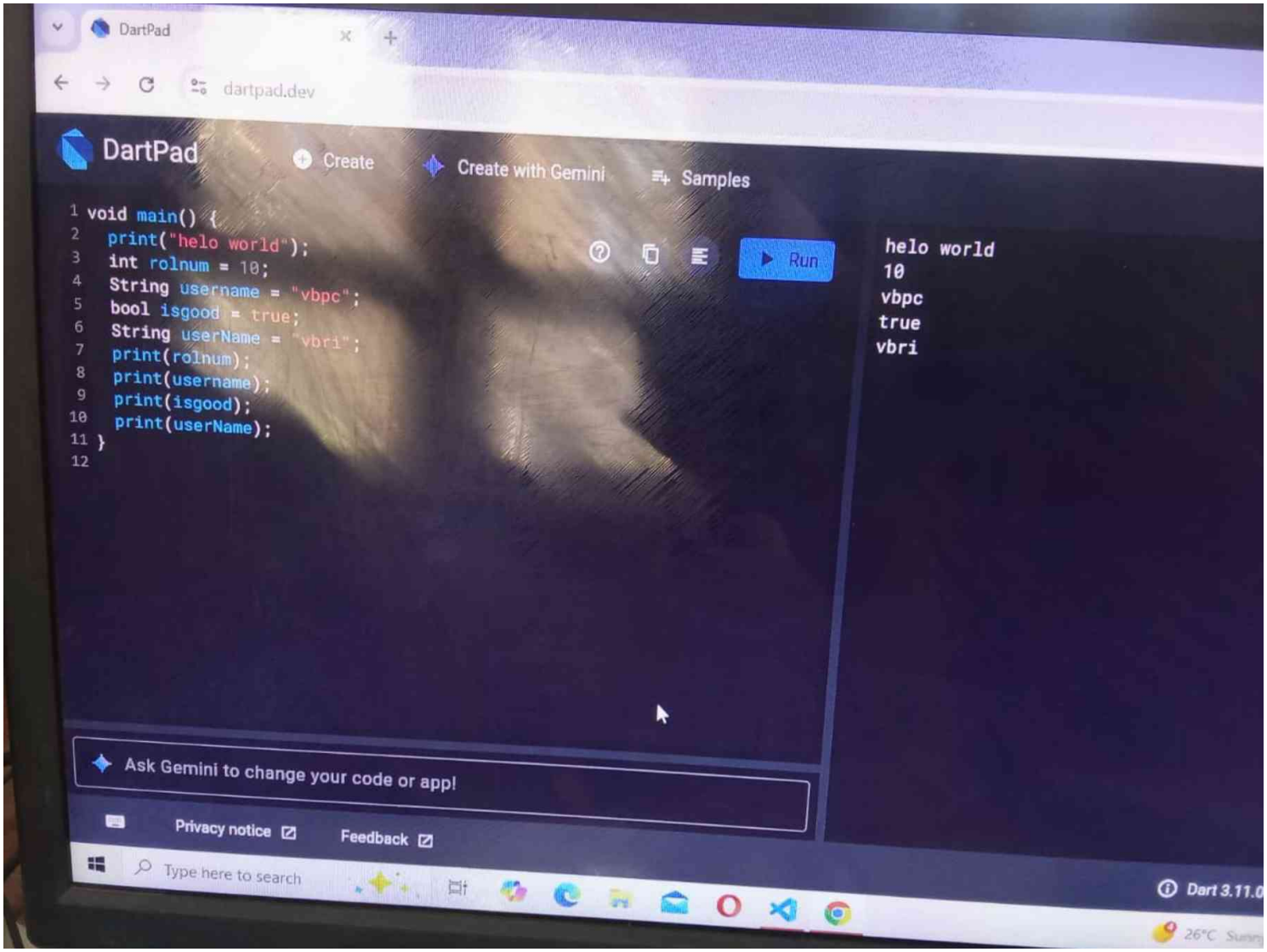
footer

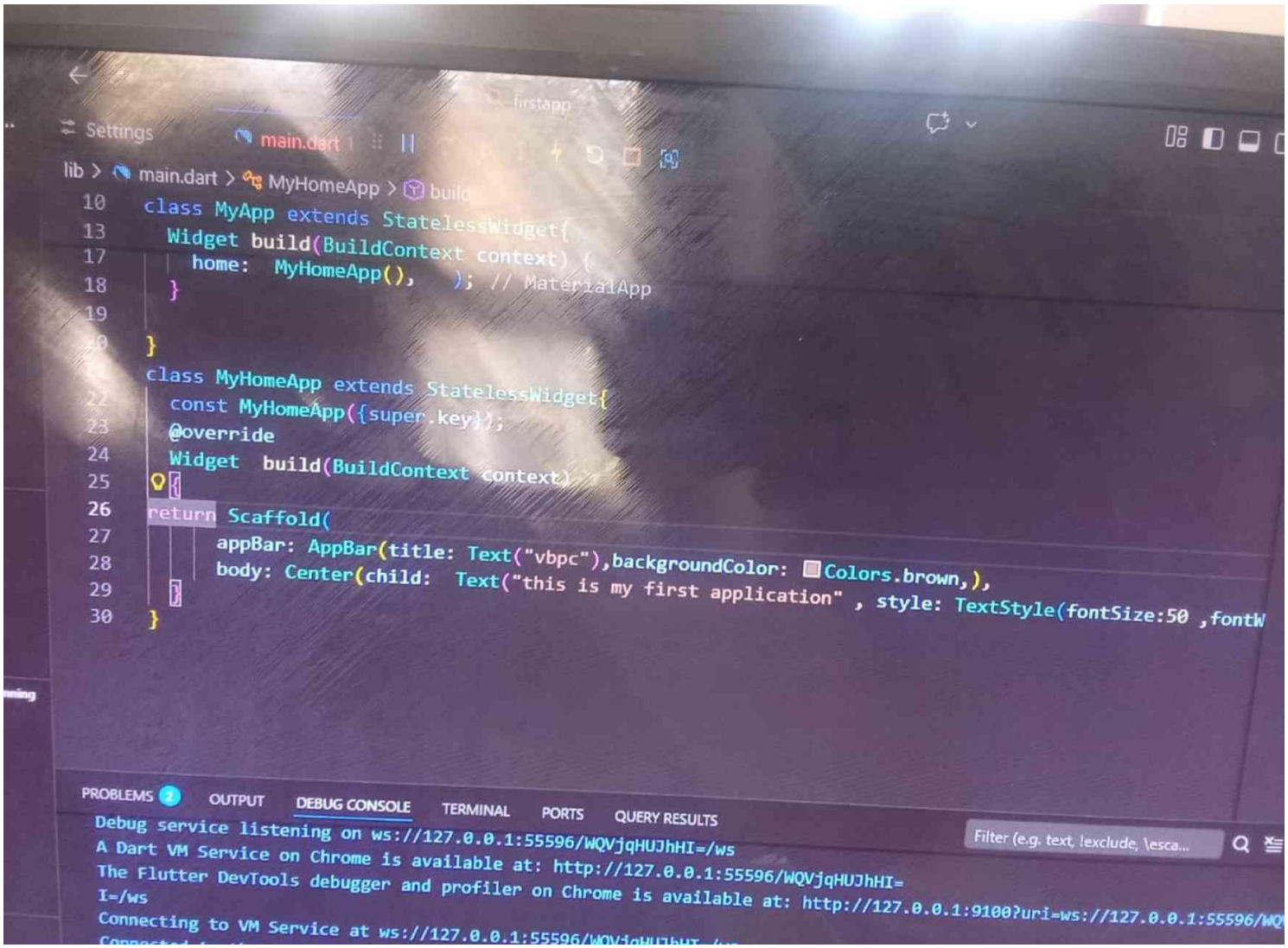
type name = value

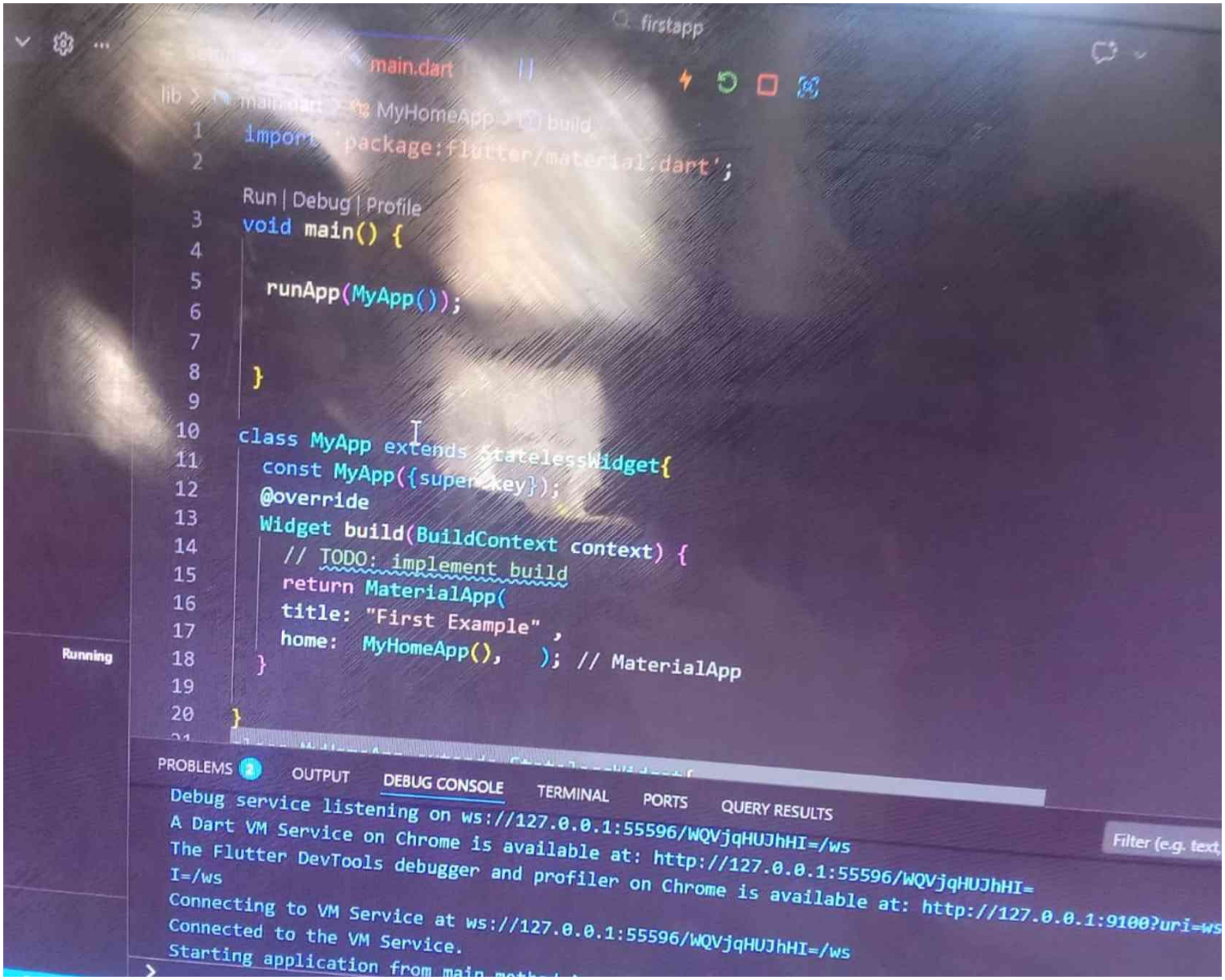
Data type		
Numbers	int, double, num, BigInt	Numbers
Strings	String	Strings
Boolean	bool	Boolean true/False
Lists	List	ordered group of objects
Maps	Map	set of values as key-value pairs

↳ keyword → not variable
↳ digit
↳ _ start, $
↳ case sensitive

var something = _;
print (' ');


DartPad
dartpad.dev
DartPad
Create
Create with Gemini
Samples
void main() {
print("helo world");
int rolnum = 10;
String username = "vbpc";
bool isgood = true;
String userName = "vbri";
print(rolnum);
print(username);
print(isgood);
print(userName);
}
Run
helo world
10
vbpc
true
vbri
Ask Gemini to change your code or app!
Privacy notice
Feedback
Type here to search
Dart 3.11.0
26°C


firstapp
Settings
main.dart
lib > main.dart > MyHomeApp > build
class MyApp extends StatelessWidget{
Widget build(BuildContext context) {
home: MyHomeApp(),); // MaterialApp
}
}
class MyHomeApp extends StatelessWidget{
const MyHomeApp({super.key});
@override
Widget build(BuildContext context)
{
return Scaffold(
appBar: AppBar(title: Text("vbpc"),backgroundColor: Colors.brown,),
body: Center(child: Text("this is my first application" , style: TextStyle(fontSize:50 ,fontW
}
}
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
QUERY RESULTS
Filter (e.g. text, !exclude, \esca...
Debug service listening on ws://127.0.0.1:55596/WQVjqHUJhHI=/ws
A Dart VM Service on Chrome is available at: http://127.0.0.1:55596/WQVjqHUJhHI=
The Flutter DevTools debugger and profiler on Chrome is available at: http://127.0.0.1:9100?uri=ws://127.0.0.1:55596/WQ
I=/ws
Connecting to VM Service at ws://127.0.0.1:55596/WQVjqHUJhHI=/ws


firstapp
main.dart
lib > main.dart > MyHomeApp > build
import package:flutter/material.dart';
Run | Debug | Profile
void main() {
runApp(MyApp());
}
class MyApp extends StatelessWidget{
const MyApp({super.key});
@override
Widget build(BuildContext context) {
// TODO: implement build
return MaterialApp(
title: "First Example" ,
home: MyHomeApp(),); // MaterialApp
}
}
Running
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
QUERY RESULTS
Filter (e.g. text,
Debug service listening on ws://127.0.0.1:55596/WQVjqHUJhHI=/ws
A Dart VM Service on Chrome is available at: http://127.0.0.1:55596/WQVjqHUJhHI=
The Flutter DevTools debugger and profiler on Chrome is available at: http://127.0.0.1:9100?uri=ws
I=/ws
Connecting to VM Service at ws://127.0.0.1:55596/WQVjqHUJhHI=/ws
Connected to the VM Service.
Starting application from main

Conditional statement -> They are basically ~~Desion~~ Decision making in programming

① If -> use it when we want to execute some task based on a condition (T/F.)
else

~~like i~~ Syntax :->

```
if (condition)
{
    // code to be executed if
       condition is true
}
else
{
    / block of code to be executed
       when condition is false
}
```

keyword

```
map<String, int> user =
    { "age" : 23,
      "usn" : 90
    };
print( user["age"] );
```

else if → it is used when we have multiple condition to

```
void main ()
{
  int i = 2;
  if ( i%2 == 0)
  {
    print ("it is even");
  }
  else
  {
    print ("it is odd");
  }

  else if (i == 9)
  {
    print ("9 is my Relation");
  }
  if ( i%2 != 0)
  {
    print ("it is another odd")
  }
```

else

conditional operator →

condition ? exp1 : exp2 ;

must Return T/F

If condition is true

condition is false

```
void main ()
{ int a=10;
int b = 20
int result = (a>b) ? a:b;
print("The greater num is %result");
}
```

```
is 19/20
print (i%2==0) ? "it is even" : "it is odd");
```

```
print (0<5);
```

```
void main()
{
  int choice = 0
   do
   { print("\n— Main menu —");
     print("1, View profile");
     print("2 setting");
          (3, Exit);
   choice ++
   print("User selected: %d choice");

   Switch (choice)
    {                          greeting
     case 1:          opening setting
       print ("Showing profile...");
       break;
     case 2:
```

```
    }
  } while(choice!=3);
}
```

Conditional ... basically Decision making

Loop
- while
- do-while
- For

① int i = 0;
while (i<5)
{ print (num[i]);
i++ }

List <int> nums = [1, 3, 3, 3, 6, 7];

```
for (int i = 0; i <= 5; i++)
{
    print (nums[i]);
}
```

```
for (int num in nums)
    print (num);
```

do {

3 while C

Arithmetic operator	operators: Comparison operator	logical operator	Assignment op
+	==	&&	=
-	!=	||	Assigna value to a variable
*	>	!	
/	<		
%	>=		
	<=		

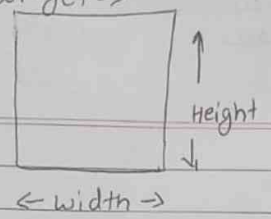
① powershell (run as admin) → choco install → choco ↵ → version
CLI → choco install -y nodejs ~~fts~~ open JDK

cmd → node -v
↳ java - version

Instal Android Studio → SDK manager
- Android SDK platform
- Sources for Android
- google play intel 8X Atom 64 system image
- Intel 86 ATOM 64 system

STD Tools ...

① Container widget →



Store →
categaric → label / color
we store widgets in container.

(SizedBox): It have three properties → Height, width, child
↳ application में space provide करता है

height: 20, width: 300,

body: center (~~Colors.amber~~, color: Colors.amber
child: Container (child: Text ("hello um", style: Text
center (child: Text (———)

layout: →

② Column widget: →

→ context
→ key
→ layout

Syntax: Column (
children: [

]
)

maxiAxis ↑
CrosAxis →

Container
↳ copy

body: Column (crossAxisAlignment: CrossAxisAlignment.Stretch,
mainAxisAlignment: MaxiAxisAlignment.center,
children: [

container
contain
contain

SizedBox (height: 20),

], // column.

Row widget ->

Syntax

Row (

children : [

]

)

mainAxis

crossAxis

Center (

child : Row

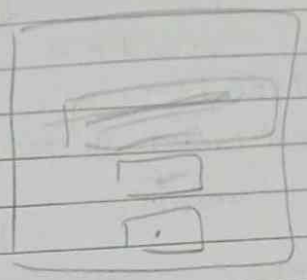
Row widget ->

Syntax

```
Row (
  children : [
  ]
)
```

maxAxis
crossAxis

Center (
child : Row



main.dart ->

```
import 'package : flutter / material . dart';

void main ()
{
    runApp( QuoteApp() );
}

class QuoteApp extends StatelessWidget
{
  const QuoteApp ({Super. key 3});

  @ override
  Widget build ( BuildContext context )
    {
        return { MaterialApp (
                title : "Random quote generator",
                home :  ,

  3 3 );
```

Statefull widget ->

```
class QuoteScreen extends StatefulWidget {
        const quoteScreen ({super.key});
    @override
    State<QuoteScreen> CreateState () => _QuoteScreenState
                                        ();
                Page                _PageState
}   //QuoteScreen -> _QuoteScreenState

class _QuoteScreenState extends State<QuoteScreen>
  {
    @override
      Widget build (BuildContext context)
        {
          return Scaffold ();
}

      backgroundColor: Colors.blueGrey[900],
  appBar: AppBar( backgroundColor: Colors.white,
      title: Text ("Random quote generator"),

        body: Column(
          children[
              Text("Push yourself, because no one else
sizedbox (height:20)          is going to do it for you"),
              Text ("unknown"),

              Button
                                    style: TexStyle
]
),                          (color: Colors.white, fontSize:24),
                                .grey[300],
                            fontStyle: FontStyle.italic ),
```